

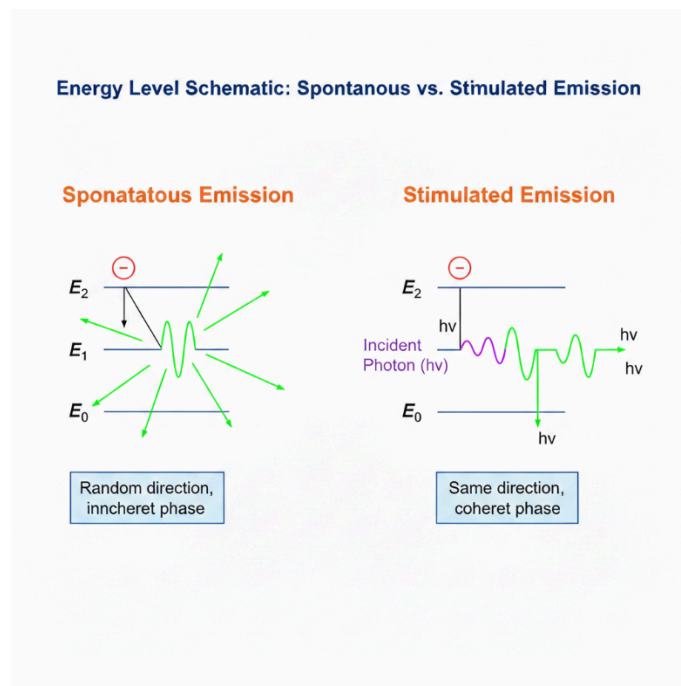
FUNDAMENTALS OF LASERS, PHOTONICS, OPTICAL FIBERS

ANSWER KEY

Q1. Explain stimulated emission vs. spontaneous emission. Draw energy level schematic (1 mark). [4 marks]

Answer: Stimulated emission occurs when a photon of energy $E_2 - E_1$ interacts with excited atom in E_2 state, forcing it to drop to E_1 state emitting identical photon (same phase, direction, wavelength). This creates coherent, amplified light for lasers. Spontaneous emission is random decay from E_2 to E_1 without external photon, producing photons with random phase/direction (incoherent light like ordinary lamps).

Energy level schematic:



Stimulated emission maintains coherence and directionality; spontaneous emission does not. Stimulated requires population inversion; spontaneous occurs naturally.

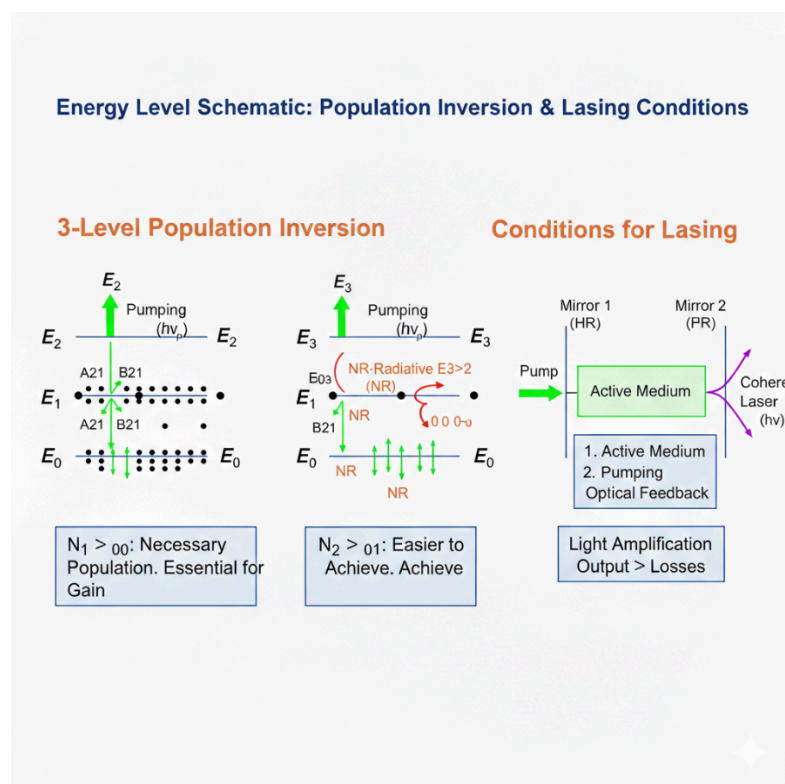
Marking: Explanation (2m) + Comparison (1m) + Diagram (1m)

Q2. Define population inversion. Explain why it's needed and optical pumping. Draw a schematic (1 mark). [4 marks].

Answer: Population inversion means more atoms in the excited state in (N_2) than in the ground state N_1 (opposite of thermal equilibrium where $N_1 \gg N_2$).

Why is it needed: Lasers require a stimulated emission rate $>$ absorption rate. Rate stim = $B_{21}N_2\rho >$ Rate abs = $B_{12}N_1\rho$ needs $N_2 > N_1$ for net gain.

Optical pumping: Intense light (shorter wavelength) excites atoms from E_1 to E_3 (higher level). Fast non-radiative decay populates E_2 , creating inversion between E_2 and E_1 . Example: Ruby laser uses a green flash lamp.



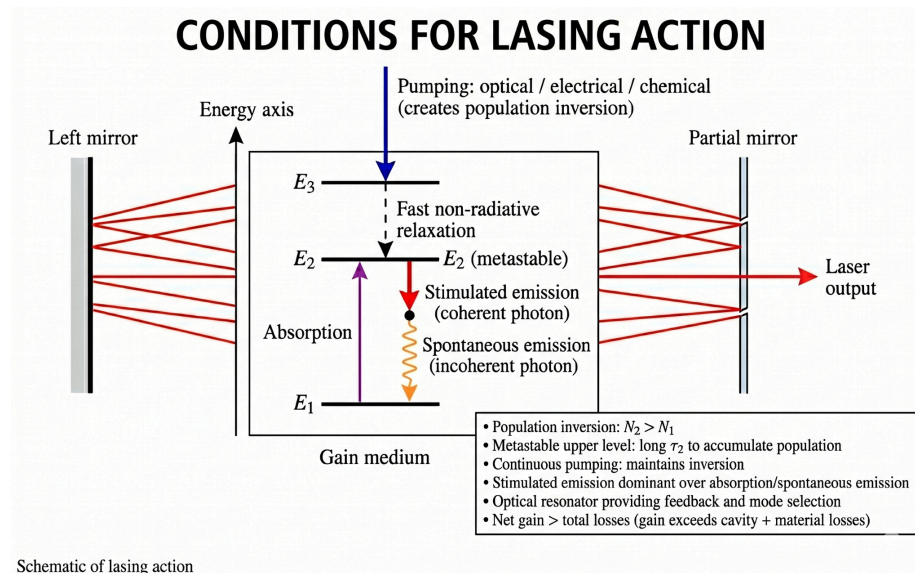
Marking: Definition (1m) + Need (1m) + Pumping (1m) + Diagram (1m)

Q3. State three conditions for laser action. Draw a basic schematic (1 mark). [4 marks]

Answer: The three essential conditions for laser action are:

1. **Population inversion** ($N_2 > N_1$) between two energy levels for net stimulated emission
2. **Optical feedback** provided by mirrors forming a resonant cavity
3. **Medium with suitable gain** that amplifies light through stimulated emission

These create self-sustaining light amplification. Cavity mirrors reflect photons back through gain medium, stimulating more emission until output coupler transmits laser beam.



Marking: Three conditions (2m) + Explanation (1m) + Diagram (1m)

Q4. Classify lasers by active medium with 1 example each. [4 marks]

Answer: Lasers are classified by active medium as:

1. **Gas lasers:** The active medium is gas/vapor. Example: He-Ne laser (red light, 632.8 nm) used in holography
2. **Solid-state lasers:** The active medium is a solid crystal doped with ions. Example: Nd:YAG laser (1064 nm) used in industry
3. **Semiconductor lasers:** Active medium is p-n junction diode. Example: Laser diode (in CD players, fiber optics)
4. **Liquid (dye) lasers:** Active medium is organic dye solution. Example: Rhodamine 6G (tunable visible range)

Marking: Classification (2m) + Examples (1m) + Applications (1m)

Q5. Explain the working of semiconductor lasers and fiber optic sensors. [4 marks]

Answer:

Semiconductor Laser: Forward-bias p-n junction injects electrons/holes, creating a population inversion across the bandgap E_g . Recombination emits photons. Cleaved facets form optical cavity mirrors. Stimulated emission between facets produces coherent output. Compact, efficient, electrically modulated. Used in CD/DVD, fiber optics.

Fiber Optic Sensors: Light propagates by TIR. Measurands modulate:

1. Intensity (bending loss)
2. Phase (interferometers)

3. Wavelength (FBG: $\Delta\lambda_B \propto \text{strain}$)

EMI immune, distributed sensing. Used in SHM, pipelines.

Marking: Semiconductor laser (2m) + Fiber sensors (2m)

Q6. Derive Einstein's coefficients (A, B) relationship for stimulated emission, absorption, and spontaneous emission. Show rate equation. [4 marks]

Answer:

Rate equations:

Absorption: $R_{12} = B_{12} \cdot N_1 \cdot \rho(\nu)$

Stimulated emission: $R_{21s} = B_{21} \cdot N_2 \cdot \rho(\nu)$

Spontaneous emission: $R_{21sp} = A_{21} \cdot N_2$

Thermal equilibrium: $R_{12} = R_{21s} + R_{21sp}$
 $B_{12} \cdot N_1 \cdot \rho = B_{21} \cdot N_2 \cdot \rho + A_{21} \cdot N_2$

Boltzmann: $\frac{N_2}{N_1} = \frac{g_2}{g_1} e^{-\frac{h\nu}{kT}}$

$$\rho(\nu) = \frac{A_{21}/B_{21}}{\frac{g_2}{g_1}(e^{h\nu/kT}-1)} = \frac{8\pi h\nu^3/c^3}{\frac{g_2}{g_1}(e^{h\nu/kT}-1)}$$

Final relations: $B_{21} = \frac{g_1}{g_2} B_{12} \frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3}$

Marking: Rates (1m) + Derivation (2m) + Final (1m)

Q7. Explain the lasing threshold condition briefly. [4 marks]

Answer:

Lasing threshold: cavity gain = total losses

$$g_{th}L = \alpha L + \ln\left(\frac{1}{\sqrt{R_1 R_2}}\right)g_{th}L = \alpha L + \ln 1/R_1 R_2$$

g_{th} = threshold gain coefficient

α = internal loss, $R_1 R_2$

L = cavity length

Below threshold: spontaneous emission only

At threshold: gain equals loss, oscillation starts

Above threshold: output $\propto$ (gain - loss)

Population inversion reaches N_{2th} where stimulated emission overcomes losses.

Marking: Condition (1m) + Equation (1m) + Explanation (2m)

Q8. List and explain 4 properties of LASER light (monochromaticity, coherence, directionality, high intensity). [4 marks]

Answer:

- Monochromaticity — unit: m (or nm).
 - Definition: Nearly single wavelength so photons have the same energy/frequency.
 - Typical scale: spectral width $\ll$ 1 nm (example: some lasers approach $\lambda \approx 0.1$ nm).

- Application: spectroscopy, holography.
- Coherence — unit: m (coherence length $l_c = c \cdot \tau_c$).
 - Definition: Fixed phase relationship in time and space (high temporal and spatial coherence).
 - Typical scale: coherence length l_c can be very long for stabilized lasers (up to km).
 - Application: interferometry, holography.
- Directionality — unit: radians (divergence θ).
 - Definition: Highly collimated beam with very low divergence, so it travels long distances with little spreading.
 - Typical scale: θ often $\ll 1$ mrad for many lasers.
 - Application: optical communication, alignment, rangefinding.
- High intensity (brightness) — units: W (power), $\text{W} \cdot \text{m}^{-2}$ (irradiance).
 - Definition: Power concentrated into a small area and narrow solid angle, producing very high irradiance and brightness.
 - Typical scale: can be $\sim 10^6$ times brighter than ordinary sources for certain lasers.
 - Application: cutting, welding, laser surgery, nonlinear optics.

Marking: 4 properties (2m) + Explanation/applications (2m)

Q9. Explain light propagation in optical fibers. Compare single vs. multimode. Draw a schematic (1 mark). [4 marks]

Answer:

Light propagates by total internal reflection (TIR) at core-cladding interface where $n_{\text{core}} > n_{\text{cladding}}$. Light ray angle $\theta < \theta_i < \text{acceptance angle}$ propagates; others leak out.

Single-mode: Core $\sim 8\mu\text{m}$, one propagation mode. Low dispersion, high bandwidth (100 km). Telecom use.

Multimode: Core $\sim 50\text{--}62.5\mu\text{m}$, multiple modes/paths. Modal dispersion limits distance (~ 2 km). LAN use.

Schematic:

Transmission of light, based on above principle, through an optical fibre of core refractive index n_1 and cladding refractive index n_2 with $n_1 > n_2$ is shown in Fig.15.3(a).

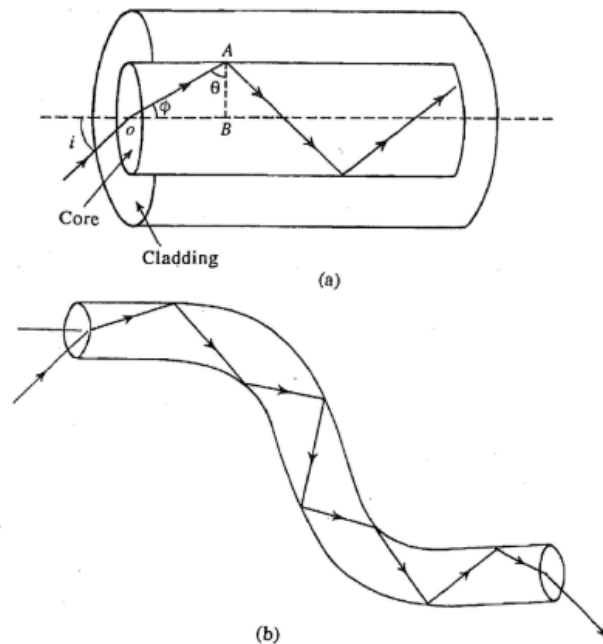
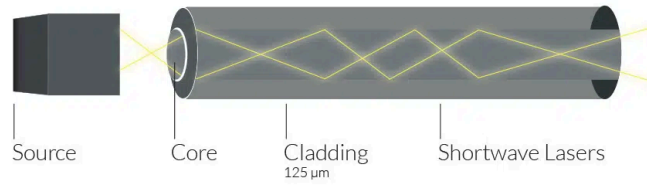


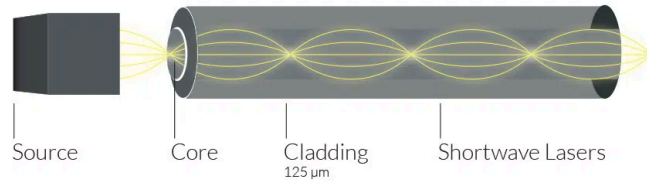
Fig.15.3: (a) Light propagation through a fibre by total internal reflection.
(b) Light propagation through a bent fibre.

Marking: Propagation (1m) + Comparison (2m) + Diagram (1m)

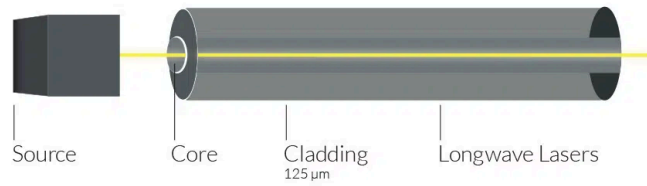
Multimode Step Index



Multimode Grade Index



Singlemode Step Index



Q10. Derive the TIR condition for fibers. Draw light path (1 mark). [4 marks]

Answer:

- Derivation Setup and Critical Condition:

- **Setup:** Ray-trace analysis of a step-index optical fiber using geometric optics and Snell's Law ($n_1 \sin \theta_1 = n_2 \sin \theta_2$).
- **Core (n_1) and Cladding (n_2):** $n_2 < n_1$.
- **Critical Acceptance Angle (θ_a):** To find θ_a , we define the Numerical Aperture (NA), which sets the maximum external acceptance angle for light to be guided. Let external medium be air ($n_0 \approx 1$).
- **Total Internal Reflection (TIR):** For a
- **Critical Angle (θ_c):** By definition at TIR

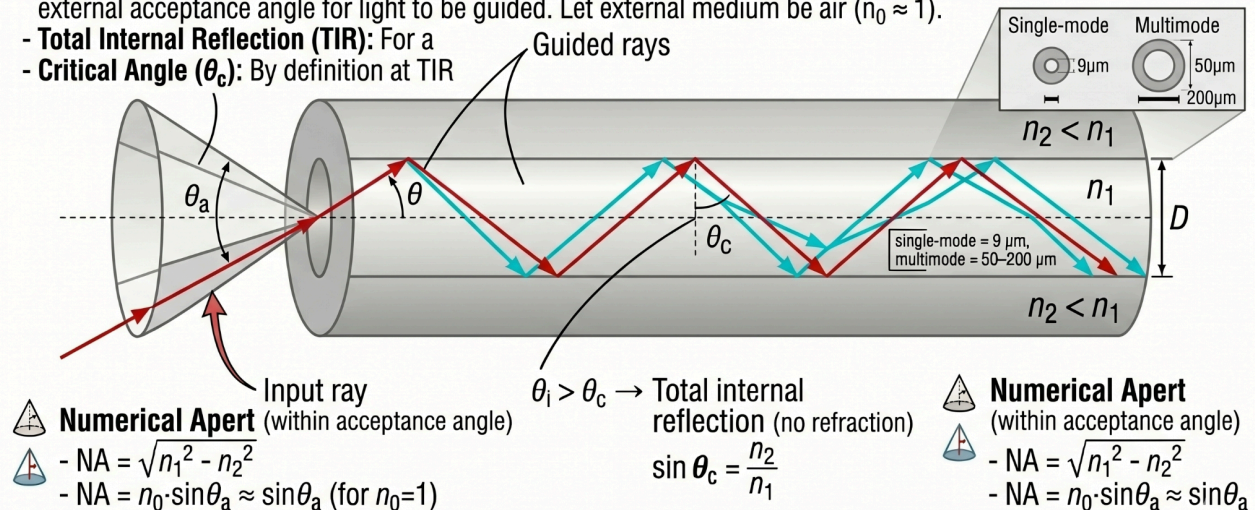


Fig. 3: Optical fiber — Total Internal Reflection

Q10. Derive the TIR condition for fibers. Draw light path. (4 marks)

Answer:

Consider a step-index optical fiber with core refractive index n_1 and cladding refractive index n_2 ($n_2 < n_1$). Let θ_i be the angle of incidence inside the core at the core-cladding interface, θ_t the transmitted angle in the cladding, θ_c the critical angle, and θ_a the acceptance angle at the input face. (Angles in rad or $^\circ$; refractive indices dimensionless.)

Derivation (3 marks)

1. Snell's law (0.5 mark)

Apply Snell's law at the core-cladding interface:

$$n_1 \sin \theta_i = n_2 \sin \theta_t.$$

(Here n are dimensionless and θ in rad or degrees.)

2. Critical angle (1 mark)

For total internal reflection take the transmitted ray grazing the interface, i.e.

$\theta_t = 90^\circ$ so $\sin\theta_t = 1$. Then

$$n_1 \sin\theta_c = n_2.$$

Hence the critical angle is

$$\theta_c = \sin^{-1}(n_2/n_1).$$

(θ_c in rad or $^\circ$.) This is the smallest internal incidence angle for which no refracted ray propagates into the cladding.

3. TIR condition and interpretation (0.5 mark)

Total internal reflection occurs when

$$\theta_i > \theta_c,$$

equivalently $n_1 \sin\theta_i > n_2$. When this holds each encounter at the interface reflects fully back into the core, so the ray is guided.

4. Numerical aperture / acceptance (1 mark)

For an external medium of refractive index n_0 (usually ≈ 1) the acceptance is defined by

$$NA = n_0 \sin\theta_a.$$

For a step-index fiber this becomes

$$NA = \sqrt{n_1^2 - n_2^2}.$$

NA (dimensionless) gives the maximum external input angle θ_a that will couple into guided rays whose internal θ_i can exceed θ_c .

Light-path summary (to include after derivation)

Input ray (within acceptance cone) $\rightarrow$ inside core: if $\theta_i > \theta_c \rightarrow$ repeated TIR at core-cladding interfaces $\rightarrow$ guided propagation; otherwise the ray is leaky and lost.

(Write neatly: Preface the figure, present the formulas and short descriptions, add the one-line light-path summary, and sketch the labeled diagram to obtain full 4 marks.)

Marking: Derivation (2m) + NA (1m) + Diagram (1m)

Q11. Draw optical fiber structure. Explain numerical aperture. Schematic (1 mark). [4 marks]

Answer:

Structure: Core (n_1 , light guide) $\rightarrow$ Cladding ($n_2 < n_1$, TIR) $\rightarrow$ Buffer jacket (protection)

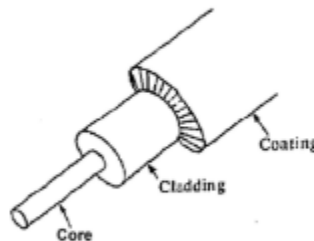


Fig. 15.1: Optical Fibre

Numerical aperture: $NA = n_0 \sin \theta_a = \sqrt{n_1^2 - n_2^2}$ $NA = n_0 \sin \theta_a = n_1^2 - n_2^2$

θ_a = max acceptance angle in air ($n_0=1$)

Higher NA = larger light cone = shorter transmission distance

Schematic: Jacket | Cladding(n_2) | Core(n_1) | Input cone (θ_a)

Marking: *Structure (1m) + NA explanation (2m) + Diagram (1m)*

Q12. Explain analogy of photon as message carrier in optical Fibers.. [4 marks]

Answer:

Photon Message Carrier - 4 Steps:

1. **Message Creation:** Electrical data $\rightarrow$ laser/LED ON/OFF modulation
(1=photon burst, 0=no photon)
2. **Photon Messenger:** Trillions of photons/second carry bits via presence/absence
3. **Highway System:** Fiber TIR guides photons 100s km (0.2 dB/km loss, Tbps capacity)
4. **Message Delivery:** Photodetector converts photons $\rightarrow$ original electrical signal

Schematic: Laser (1/0) $\rightarrow$ Photons $\rightarrow$ Fiber(TIR) $\rightarrow$ Detector $\rightarrow$ Data

Marking: *4 steps (2m) + Explanation (1m) + Diagram (1m)*

Q13. List 3 main fiber losses and 1 reduction method each. [4 marks]

Answer:

1. **Absorption loss** (material): Use pure silica, low-OH content fibers
2. **Rayleigh scattering** (density fluctuations): Minimize core/cladding imperfections
3. **Bending loss** (macro bend): Maintain minimum bend radius $> 30\text{mm}$

Marking: 3 losses (2m) + 3 methods (2m)

Q14. Compare step-index vs. graded-index fibers. Draw both (1 mark). [4 marks]

Answer:

Step-index: Sharp $n_1 \rightarrow n_2$ step | Straight/meridional rays | High modal dispersion | Bandwidth $\sim 500\text{ MHz-km}$

Graded-index: Parabolic $n(r)$ profile | Helical rays equalize path lengths | Low dispersion | Bandwidth $\sim 2\text{ GHz-km}$

Schematics:

Step: Core(n_1) || Cladding(n_2) -- straight rays

Graded: Core($n_1 \text{ max} \rightarrow n_2 \text{ min}$) -- helical paths

Marking: Comparison (2m) + Characteristics (1m) + Diagrams (1m)

Q15. State 2 optical fiber applications. Draw 1 sensor schematic (1 mark). [4 marks]

Answer:

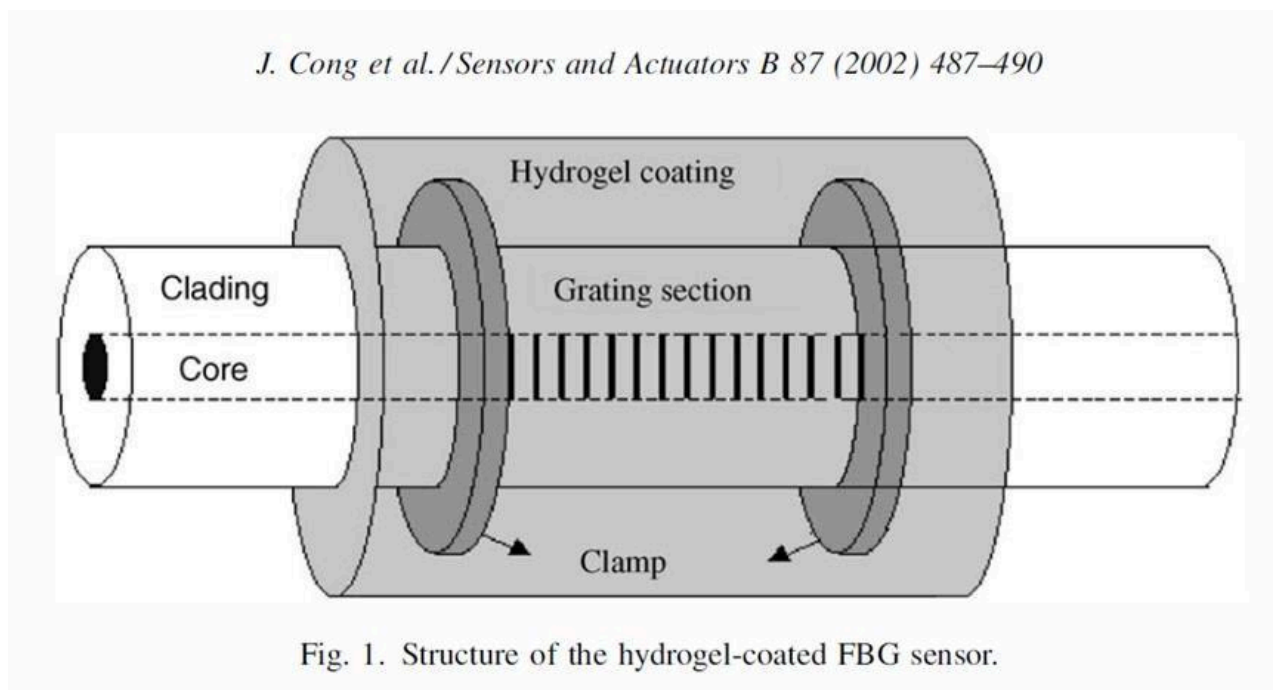
1. **Telecommunications:** Single-mode 1550nm Fibers (Transoceanic cables)
2. **Sensors:** Fiber Bragg Grating (FBG) - strain/temp monitoring

FBG (Fiber Bragg Grating) Sensor: Periodic RI grating reflects $\lambda_B = 2n_{eff}\Lambda$

$$\lambda_B = 2n_{eff}\Lambda$$

Strain $\rightarrow \Delta\Lambda \rightarrow \Delta\lambda_B$ measured by spectrometer

Schematic:



Marking: Applications (2m) + FBG principle (1m) + Diagram (1m)